

TRAINING PHASE

Prior
 $p(\theta)$

sample θ

Simulator
 $p(Y|\theta)$

Simulate N times

Simulated
Data
 $\{(\theta_i, Y_i)\}_{i=1}^N$

Train Network

Neural
Density
Estimator
 $q_\phi(\theta|Y)$

Evaluate

Posterior
 $q(\theta|Y_{\text{obs}})$

INFERENCE PHASE

Observed
Data
 Y_{obs}